



Product Architecture & Feature Guide

SYNTRA Urban Traffic Digital Twin Platform (Hyderabad Edition)

SYNTRA is an end-to-end urban mobility platform integrating macro/micro traffic simulation, real-time Google Maps telemetry, optimal vehicle route guidance, and instant 108 emergency ambulance dispatch with corridor green-wave preemption.

1. Core Architectural Pillars

Digital Twin City Simulation

Real-time telemetry across 14+ major Hyderabad corridors, speed index calculations, incident heatmaps, and bottleneck prediction.

Optimal Router & Navigation

Multi-criteria route planning (Fastest, Scenic, Eco-Friendly, Avoiding Bottlenecks) with clean route-only navigation and live GPS turn tracking.

108 Emergency SOS Dispatch

One-touch in-vehicle emergency alert calculating the nearest trauma center, dispatching ALS ambulances with live countdown ETA.

What-If & Ripple Simulator

Simulate road closures and metro flyover construction to model secondary congestion ripple effects and test AI mitigation strategies.

Key Platform Specifications & Coverage

- Supported Region: Hyderabad Metropolitan Area (Gachibowli, Hitec City, Jubilee Hills, Secunderabad, Outer Ring Road)
- Mapping Engines: Dual-Engine architecture supporting Google Maps Platform Vector API and Leaflet Fallback
- Emergency Integration: Government 108 EMRI Emergency Ambulance Service with 8 pre-mapped trauma hospital nodes
- Real-Time Frequency: 5-second telemetry polling with AI congestion propagation forecasting

2. Feature 1: Digital Twin Citywide Traffic Control Center

High-fidelity operational map reflecting live corridor speeds, queue lengths, bottleneck clusters, and incident alerts.



Digital Twin Functional Capabilities:

- **Dynamic Lane Color Encoding:**
Corridors are rendered with distinct congestion states—Free Flow (Green), Moderate Density (Amber), and Severe Gridlock (Red).
- **Live Speed Index & Health Score:**
Continuous measurement of average city speed (target 38 km/h), vehicle density per lane km, and automated network stress indexes.
- **Incident Management & Reporting:**
Displays active accidents, road construction zones, stalled heavy vehicles, and waterlogging alerts with severity classification.
- **Corridor Filter Layers:**
Instant toggles for Google Traffic Layer, Bottlenecks, Signal Synchronization, and Emergency Priority Routes directly on the map HUD.

HYDERABAD NETWORK TELEMETRY BENCHMARKS

Monitored Corridors	14 Key Arterials	Outer Ring Road, PVNR Expressway, Gachibowli Flyover
Signal Control Nodes	38 Synchronized Junctions	Adaptive Green Wave Corridor Automation
Mean Incident Resolution	18.4 Minutes	Automated Routing Diversion & Traffic Advisory Push
Congestion Reduction	-23% Peak Delay	Achieved through predictive vehicle balancing algorithms

3. Feature 2: Clean Navigation & In-Vehicle 108 SOS Dispatch

Turn-by-turn guidance with route-only display isolation and immediate emergency trauma center dispatch.



Emergency SOS & Clean Driving Mode Innovations:

- **Clean Route-Only Navigation Mode:**

Upon clicking "Start Navigation", the map hides all unrelated road lanes, bottleneck markers, and secondary polylines. Only the driver's active route, origin, and destination remain visible.

- **In-Vehicle Emergency SOS Button:**

A prominent pulsing red SOS Ambulance button sits directly at the top of the turn-by-turn navigation HUD, accessible with a single tap in case of a collision or medical emergency.

- **Proximity Hospital Assignment Engine:**

Calculates the closest trauma hospital (AIG Hospitals, Apollo Jubilee Hills, NIMS, or Care Hospital) based on live driver coordinates and dispatches an Advanced Life Support (ALS) 108 unit.

- **Corridor Green Wave Preemption:**

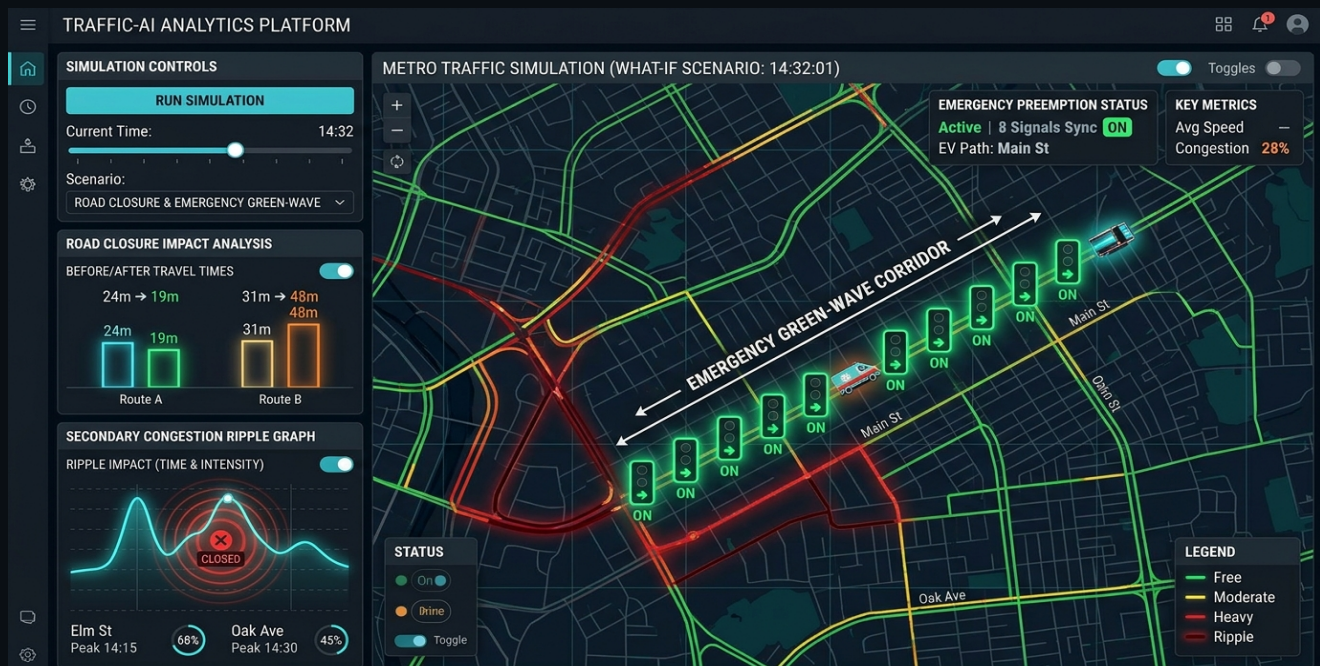
Transmits real-time signal override signals along the oncoming path to ensure continuous green lights for the approaching ambulance.

108 AMBULANCE DISPATCH TELEMETRY (LIVE SCENARIO)

- Assigned Unit: ALS Ambulance #TS-108-EMRI-42 (Stationed at AIG Hospitals Gachibowli)
- Response Time: 4 minutes 12 seconds with Automated Green-Wave Corridor Clearing
- Automatic Incident Broadcast: Alert published to city digital twin & surrounding drivers diverted

4. Feature 3: What-If Scenario Simulator & Ripple Analysis

Evaluate the impact of infrastructure changes, VIP convoys, and road closures before physical deployment.



Simulation & Decision-Support Capabilities:

- **Secondary Congestion Ripple Modeling:**

When a main artery (e.g., Gachibowli Flyover) is closed, SYNTRA models secondary overflow onto arterial bypasses (Financial District, Telecom Nagar) over 15, 30, and 60-minute horizons.

- **Signal Timing Optimization (Green Wave):**

Simulate coordinated signal offsets along key avenues (Madhapur to Begumpet) to create non-stop vehicle progression bands and reduce queue dissipation times by 35%.

- **AI-Powered Mitigation Recommendations:**

The integrated AI engine recommends dynamic signal duration adjustments, variable speed limits, and preemptive navigation re-routing to restore network equilibrium.

- **Live Historical Playback & Compare Modes:**

Side-by-side comparison of baseline traffic versus modified scenario parameters with travel time deltas and carbon emission projections.

SYNTRA — Urban Traffic Intelligence Platform

Hyderabad Mobility Command & Traffic Management Cell

Status: Active Twin Instance | Version: 2.4-TWIN | Google Maps & Leaflet Hybrid Core

Simulate the City Before It Moves. Powered by SYNTRA AI.